Statistical Inference for Linear Stochastic Approximation with Richardson–Romberg Extrapolation

Danil Gainanov

Supervisor: Ilya Levin

Master's thesis defense, 2026

Motivation: What the Recursion Solves

Linear stochastic approximation (LSA) drives **TD-learning and policy evaluation** in reinforcement learning. From a **dependent data stream** $Z_1, Z_2, \dots$ (a Markov chain), the constant-step recursion

$$\theta_k^{(\alpha)} = \theta_{k-1}^{(\alpha)} - \alpha\big(A(Z_k)\theta_{k-1}^{(\alpha)} - b(Z_k)\big)$$

solves the mean linear system

$$\overline{A}\theta^* = \overline{b}, \qquad \overline{A} = \mathbb{E}_\pi[A(Z)], \qquad \overline{b} = \mathbb{E}_\pi[b(Z)], \qquad \theta^* = \overline{A}^{-1}\overline{b}.$$

Canonical instance: TD(0).
Features $\phi(s)$, $V(s) \approx \phi(s)^\top\theta$, transition $Z_k = (s_k, r_k, s_{k+1})$:

$$A(Z_k) = \phi(s_k)\big(\phi(s_k) - \gamma\phi(s_{k+1})\big)^\top, \qquad b(Z_k) = r_k\phi(s_k);$$

θ^* is the TD fixed point.

Why inference is hard.

- Constant step does not converge to θ^*: a steady-state **bias of order α**.
- A biased center invalidates a CI; no variance estimate repairs a wrong center.

Goal: a valid **confidence interval** for $u^\top\theta^*$—RR corrects the center, long-run variance sets the width.

Plan: (1) setup & target; (2) Richardson–Romberg & the main Berry–Esseen bound; (3) proof architecture; (4) confidence intervals & experiments.

Setup and Assumptions

Centering the recursion at θ^* gives the **error recursion**

$$\theta_k^{(\alpha)} - \theta^* = \big(I - \alpha A(Z_k)\big)\big(\theta_{k-1}^{(\alpha)} - \theta^*\big) - \alpha\varepsilon(Z_k), \qquad \varepsilon(z) = \widetilde{A}(z)\theta^* - \widetilde{b}(z).$$

with $\widetilde{A} = A - \overline{A}$, $\widetilde{b} = b - \overline{b}$, so the noise is centered: $\pi(\varepsilon) = 0$.

Assumptions

- **(A1)** Uniform geometric ergodicity, mixing time t_{mix}.
- **(A2)** $-\overline{A}$ Hurwitz; A, $\widetilde{A}$ bounded.
- **(A3)** Bounded noise $\|\varepsilon\|_\infty < \infty$.

Polyak–Ruppert average

Burn-in n_0, window $m = n - n_0$:

$$\overline{\theta}_{n,n_0}^{(\alpha)} = \frac{1}{m}\sum_{k=n_0}^{n-1} \theta_k^{(\alpha)}.$$

Markovian noise means the randomness in $A(Z_k)$, $b(Z_k)$ is serially dependent; the constant step α does not vanish, so the iterates have a steady-state bias.

Statistical Target and Two Error Sources

For confidence intervals we need a normal approximation for scalar projections

$$\sqrt{n}\, u^\top(\bar{\theta}_n - \theta^*) \approx \mathcal{N}(0, u^\top \Sigma_\infty u).$$

Two distinct obstacles:

Centering error

$$\mathbb{E}\,\bar{\theta}_n^{(\alpha)} - \theta^* = \alpha\Delta + \text{higher order}.$$

A variance estimator cannot correct a biased interval center.

Dependent fluctuation

$$\Sigma_\varepsilon^{\mathrm{M}} = \mathbb{E}[\varepsilon_0\varepsilon_0^\top] + 2\sum_{\ell\geq 1}\mathbb{E}[\varepsilon_0\varepsilon_\ell^\top].$$

The normal variance must include serial correlations.

Two separate tasks: **correct the center** of the estimator, and **approximate the dependent-noise distribution** around that center. RR handles the first.

Richardson–Romberg in the Step Size

Run two PR-averaged LSA trajectories on the **same Markov chain path**, with steps α and 2α, and extrapolate:

$$\bar{\theta}_n^{\mathrm{RR}} = 2\bar{\theta}_n^{(\alpha)} - \bar{\theta}_n^{(2\alpha)}.$$

Both branches carry the same leading bias slope Δ (Levin et al., 2025), so the linear term cancels:

$$\bar{\theta}_n^{(\alpha)} \approx \theta^* + \alpha\Delta, \qquad \bar{\theta}_n^{(2\alpha)} \approx \theta^* + 2\alpha\Delta \quad \Longrightarrow \quad \bar{\theta}_n^{\mathrm{RR}} \approx \theta^* + (2\alpha - 2\alpha)\Delta = \theta^* + O(\alpha^{3/2}).$$

RR moves the **center** of the confidence interval; it does not estimate the covariance. The shared path keeps the two branches' noise highly correlated, so extrapolation cancels bias instead of adding simulation error.

What Distribution Should We Approximate?

For a direction u and window $m = n - n_0$, the practical statistic is

$$\sqrt{m}\, u^\top(\overline{\theta}^{\mathrm{RR}}_{n,n_0} - \theta^*) \quad \Longrightarrow \quad \mathcal{N}(0, \sigma^2(u)), \qquad \sigma^2(u) = u^\top \Sigma_\infty u.$$

Because the data are dependent, the target is the **long-run** covariance, not a one-step variance:

$$\Sigma_\infty = \overline{A}^{-1} \Sigma_\varepsilon^{\mathrm{M}} \overline{A}^{-\top},$$

$$\Sigma_\varepsilon^{\mathrm{M}} = \mathbb{E}_\pi[\varepsilon_0 \varepsilon_0^\top] + \sum_{j \geq 1} \left(\mathbb{E}_\pi[\varepsilon_0 \varepsilon_j^\top] + \mathbb{E}_\pi[\varepsilon_j \varepsilon_0^\top] \right).$$

The inference goal is a Gaussian approximation of the scalar RR statistic with the Markovian long-run variance $\sigma^2(u) = u^\top \Sigma_\infty u$.

Main Theoretical Result

Under (A1)–(A3), at the **balanced scale** $\alpha = cn^{-1/2}$ with burn-in $n_0 \asymp (\alpha a)^{-1} \log^2 n$—where $c > 0$ is the step-size constant and a the Lyapunov contraction rate—the burned-in RR average satisfies a Berry–Esseen bound:

$$d_{\mathrm{K}}\left(\frac{\sqrt{m}\, u^\top(\bar{\theta}^{\mathrm{RR}}_{n,n_0} - \theta^*)}{\sigma(u)}, \mathcal{N}(0,1)\right) \le C(u, c, \theta_0)\ \mathrm{polylog}(n)\, n^{-1/4}, \qquad \sigma^2(u) = u^\top \Sigma_\infty u.$$

The theorem turns the RR averaged estimator into a valid normal-approximation object for scalar confidence intervals.

The constant $C(u, c, \theta_0)$ depends only on the direction u, the step-size constant c in $\alpha = cn^{-1/2}$, and the start θ_0. The $n^{-1/4}$ rate is the martingale Berry–Esseen rate; polylog factors absorb the mixing time and the burn-in.

Theorem 1 (1/3): Decomposition and RR Weights

In the stationary augmented chain ($n_0 = 0$), unfolding the error recursion and PR-averaging splits the scaled error into a noise sum plus deterministic remainders:

$$\sqrt{n}\,(\overline{\theta}_n^{\mathrm{RR}} - \theta^*) = W^{\mathrm{RR}} + D^{\mathrm{RR}}, \qquad W^{\mathrm{RR}} = -\frac{1}{\sqrt{n}} \sum_{\ell=1}^{n-1} \mathcal{Q}_\ell^{\mathrm{RR}}\, \varepsilon(Z_\ell).$$

The deterministic **RR weight** is $\mathcal{Q}_\ell^{\mathrm{RR}} = 2Q_\ell^{(\alpha)} - Q_\ell^{(2\alpha)}$, with $Q_\ell^{(\alpha)} = \overline{A}^{-1}(I - B_\alpha^{n-\ell})$ and $B_\alpha = I - \alpha\overline{A}$ (write $k = n - \ell$):

Closeness to $\overline{A}^{-1}$

$$\mathcal{Q}_\ell^{\mathrm{RR}} - \overline{A}^{-1} = -\overline{A}^{-1}(2B_\alpha^k - B_{2\alpha}^k),$$

$$\sum_\ell \|\mathcal{Q}_\ell^{\mathrm{RR}} - \overline{A}^{-1}\|^2 \le \frac{C_1}{\alpha a}.$$

Bounded variation

$$\sum_\ell \|\mathcal{Q}_{\ell+1}^{\mathrm{RR}} - \mathcal{Q}_\ell^{\mathrm{RR}}\| \le \frac{C_2}{a^2}.$$

Controls the Abel / Poisson boundary remainder.

All stochastic content of the leading term sits in $\varepsilon(Z_\ell)$; the kernel is deterministic and asymptotically $\overline{A}^{-1}$.

Theorem 1 (2/3): Poisson Martingale and Variance

W^{RR} is a dependent sum. Solve the **Poisson equation** $\widehat{\varepsilon} - \mathsf{Q}\widehat{\varepsilon} = \varepsilon$ and split each $\varepsilon(Z_\ell)$ into a martingale increment plus a telescoping term; Abel summation gives

$$W^{\mathrm{RR}} = -\frac{1}{\sqrt{n}} M_n^{\mathrm{RR}} + D_{2,n}^{\mathrm{RR}}, \qquad \Delta M_\ell^{\mathrm{RR}} = \mathcal{Q}_\ell^{\mathrm{RR}}\big(\widehat{\varepsilon}(Z_\ell) - \mathsf{Q}\widehat{\varepsilon}(Z_{\ell-1})\big),$$

where the Abel remainder is $\|D_{2,n}^{\mathrm{RR}}\|_{L^p} = O(t_{\mathrm{mix}} n^{-1/2})$.

Variance matches the target

$$\Sigma_n^{\mathrm{RR}} = \frac{1}{n}\sum_{\ell=2}^{n-1} \mathcal{Q}_\ell^{\mathrm{RR}} \Sigma_\varepsilon^{\mathrm{M}} (\mathcal{Q}_\ell^{\mathrm{RR}})^\top,$$

$$\|\Sigma_n^{\mathrm{RR}} - \Sigma_\infty\| \le \frac{C_3}{n\alpha a}.$$

Martingale Berry–Esseen

$$d_{\mathrm{K}}\left(\frac{u^\top M_n^{\mathrm{RR}}}{\sqrt{n}\,\sigma_n^{\mathrm{RR}}(u)}, \mathcal{N}(0,1)\right) \le \frac{C\log^{3/4} n}{n^{1/4}} + \frac{C\log n}{\sqrt{n}}.$$

The dependent fluctuation reduces to a bounded-increment martingale whose variance is Σ_∞ up to $O(1/(n\alpha a))$.

Theorem 1 (3/3): Misadjustment and Assembly

The remaining non-martingale piece is the RR **misadjustment**

$$R_n^{\mathrm{mis,RR}} = \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} \big(2R_k^{(\alpha)} - R_k^{(2\alpha)}\big).$$

A depth-two refinement of the perturbation recursions (Levin et al., 2025) controls it at the same order as the martingale rate:

$$\|R_n^{\mathrm{mis,RR}}\|_{L^p} \le C \ \mathrm{polylog}(n)\, n^{-1/4} \qquad \text{at} \qquad \alpha = cn^{-1/2}.$$

A Bobkov–Götze **smoothing inequality** then merges the martingale Berry–Esseen term and this remainder into a single Kolmogorov bound,

$$d_{\mathrm{K}}(X_n + Y_n, \cdot) \le d_{\mathrm{K}}(X_n, \cdot) + \frac{e\|Y_n\|_{L^p}}{\sqrt{2\pi}} + e^{-p}.$$

Stationary balanced-scale bound:

$$d_{\mathrm{K}}\left(\frac{S_{n,\mathrm{stat}}^{\mathrm{RR}}(u)}{\sigma_n^{\mathrm{RR}}(u)}, \mathcal{N}(0,1)\right) \le C(u)\,\mathrm{polylog}(n) n^{-1/4},$$

with covariance target Σ_∞.

Theorem II: Burn-in Transfer to Deterministic Starts

The real algorithm starts from a fixed θ_0 and a non-stationary Z_0. We transfer the stationary bound to the burned-in statistic

$$T_{n,n_0}^{\mathrm{RR}}(u) = \sqrt{m}\, u^\top (\bar{\theta}_{n,n_0}^{\mathrm{RR}} - \theta^*), \qquad m = n - n_0.$$

Four extra sources, all controlled

- deterministic transient $B_\alpha^k(\theta_0 - \theta^*)$;
- random initial products $\Gamma_{1:k}^{(\alpha)} - B_\alpha^k$;
- augmented-chain startup discrepancy;
- finite-window variance normalization.

Balanced burn-in result

With $n_0 \asymp (\alpha a)^{-1} \log^2 n$ and $\alpha = cn^{-1/2}$:

$$d_{\mathrm{K}}\left(\Xi_{n,n_0}^{\mathrm{RR}}(u), \mathcal{N}(0,1)\right) \le C(u, c, \theta_0)\, \mathrm{polylog}(n) n^{-1/4}.$$

This is the theorem behind the estimator used in experiments: discard the burn-in, average the remaining RR iterates, and use the same covariance target Σ_∞.

In practice Σ_∞ is unknown, so the long-run variance of the projected trajectory $Y_t = u^\top \theta_t$ is estimated from one dependent run.

OBM (overlapping batch means, block b)

$$\widehat{\sigma}^2_{\mathrm{OBM}}(b) = \frac{b}{T-b+1} \sum_{s=0}^{T-b} \left(\overline{Y}_{s,b} - \overline{Y}_T\right)^2.$$

Window bias $\approx c_1(u)/b$, often negative.

OBM-LW / lugsail ($\lambda > 1$)

$$\widehat{\sigma}^2_{\mathrm{LW}} = \frac{\lambda}{\lambda-1}\widehat{\sigma}^2_{\mathrm{OBM}}(\lambda b) - \frac{1}{\lambda-1}\widehat{\sigma}^2_{\mathrm{OBM}}(b).$$

$\lambda = 2$ cancels the leading $1/b$ bias.

RR changes the center; OBM and lugsail change the width. The thesis proves the RR distributional approximation; non-asymptotic OBM/lugsail theory along RR trajectories is left as future work.

Numerical Experiments

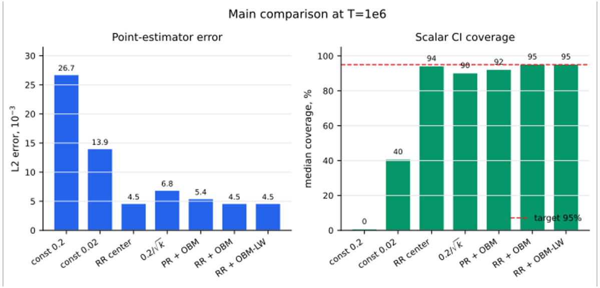
Quantity	Main setup
Problem class	Finite-state Markovian LSA
Dimension and states	$d = 5$, 10 Markov states
Monte Carlo design	100 problems, 100 trajectories per problem
Trajectory length	$T = 10^6$
RR stepsizes	$\alpha \in \{0.2, 0.02\}$ in the main comparison
Inference	95% scalar intervals in a random projection direction
Variance estimators	Batch means, OBM, OBM-LW, MSB, and oracle diagnostics

The experiments are designed to separate the point-estimator bias from the long-run variance estimation error.

Main Comparison: RR Fixes the Center

Method	L2	Cov.
$\alpha = 0.2$ const.	26.67	0.5%
$\alpha = 0.02$ const.	13.93	40.5%
RR const.	4.52	94.0%
Diminishing	6.80	90.0%
PR + OBM	5.35	92.0%
RR + OBM	4.52	95.0%
RR + OBM-LW	4.52	95.0%

L2 is reported in units of 10^{-3}; medians over 100 problems.



The coverage gain is not achieved by widening intervals: RR mainly improves the interval center by reducing step-size bias.

Diagnostics: Stepsize and Variance Estimation

RR pair	Single-step coverage	RR L2	RR cov.
$(0.04, 0.02)$	0.04: 92.0%; 0.02: 94.0%	2.97	94.0%
$(0.10, 0.05)$	0.10: 86.0%; 0.05: 91.5%	2.97	94.0%
$(0.20, 0.10)$	0.20: 67.5%; 0.10: 86.0%	2.97	94.0%

For the largest adjacent pair, oracle and practical variance intervals are almost identical:

RR + oracle variance: 95.0%; RR + OBM: 95.0%; RR + MSB: 95.0% coverage.

Across $T \in \{2 \cdot 10^4, \ldots, 10^6\}$ the oracle interval stays near 95% coverage, so the RR center and the normal approximation are not the bottleneck. The only short-horizon gap is variance estimation: at $T = 2 \cdot 10^4$ the data-driven interval is about 5.8% too narrow, and this gap closes as T grows.

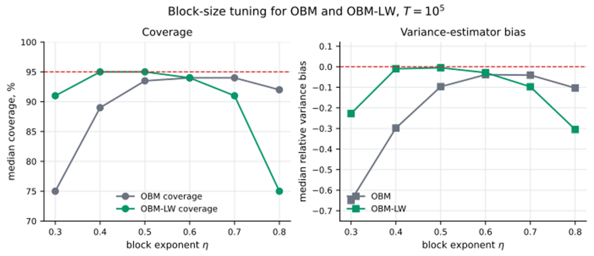
Variance Estimation: When Does Lugsail Help?

The remaining short-horizon gap is in **variance estimation**, not the center.

OBM can be too narrow when the Bartlett-window bias is negative.

Lugsail helps at small blocks / short horizons: at $T = 2 \cdot 10^4$, $\eta = 0.5$ it lifts median coverage $91.5\% \to 95.0\%$.

But not automatic: neutral at the production rule $\eta = 0.6$, and the signed estimate can turn negative for very large blocks.



Step-size RR and lugsail OBM solve different problems: center bias versus variance-estimator window bias.

Contributions and Limitations

Main contributions

- Formulated RR-averaged constant-step LSA inference under Markovian noise.
- Proved a scalar Berry–Esseen bound at the balanced scale.
- Transferred the result from stationary starts to burned-in deterministic starts.
- Verified the bias-correction mechanism in finite-state experiments.

Limitations and future work

- Full multivariate confidence regions.
- Non-asymptotic OBM and lugsail theory along RR trajectories.
- Sharper dependence on mixing time and stability constants.
- More diagnostics for slow mixing and matrix covariance estimates.

The central message: RR makes constant-step LSA statistically usable by correcting the center, while standard long-run variance tools handle the interval width.

Thank you!

Questions?